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Deep multi-task learning structure for segmentation and classification of supratentorial brain tumors in MR images

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ABSTRACT

Identification of brain tumors border and determination of their possible pathology in MR images is an important step in pre-operation analyzing of this serious medical condition. Manual segmentation and classification of brain tumors could be challenge full in neurosurgical practice because of vast differences between brain tumors characteristic such as shape, border irregularity, consistency and etc. as well as interobserver variations. To solve this problem, some automatic methods have been proposed for brain tumors segmentation or classification during recent years, but an intelligence-based method for simultaneous identification of tumor type and tumor border in MR images has not proposed till now. Here, we have planned a unique automatic model includes a common encoder for feature representation, one decoder for segmentation and a multi-layer perceptron for classification of three common primary brain tumors (meningiomas, gliomas and pituitary adenomas) in brain MR images. The proposed model was examined on a brain tumor images dataset and the output were evaluated in both multi-task and single-task learning model. The multi-task learning model gains significant improvement in simultaneous classification and segmentation of brain tumors with promising accuracy of 97% for each task. So, this model could serve as a primary screening tool for early diagnosis of common primary brain tumors in general practice with a high success rate.

1. Introduction

Brain tumors account for one every 100 cancers diagnosed annually in the United State and they can significantly impact the quality of life and change everything for the patients and their families [1–3]. Glioma are the most common primary brain neoplasma in adults and have an incidence rate of 6.5 per 100,000 in the United States. They originate in the glial cells that support neurons of the brain including astrocytes, oligodendrocytes and ependymal cells. Glioma [4–6] are classified according to the type of glial cell involved in the tumor and tumor's genetic features, which can help predict how the tumor will behave overtime and treatments most likely to work. A glioma can affect the brain function and be life-threatening depending on its location and rate of growth.

Meningiomas constitute 15% to 20% of all primary intracranial tumors in surgical series, but their incidence in routine screening is 1 in 100 population. They are believed to drive from the arachnoid cap cell near venous sinuses and exhibit a wide variety of behaviors from benign to extremely aggressive. Meningiomas present with a wide variety of symptoms, but many are asymptomatic signs and symptoms are related to compression of adjacent structures, such as seizures, motor and sensory deficits, speech disturbance, anosmia and etc. Pituitary tumors are the third most common primary intracranial neoplasma, behind glioma and meningiomas account for nearly 10% of all surgically resected primary brain tumors. They present in various ways as a result of excess or deficient secretion of pituitary hormons or extrinsic compression on the pituitary stalk or adjacent structures especially optic chiasm.

Magnetic resonance imaging (MRI) is the gold standard non-invasive diagnostic tool for brain tumors. In comparison with routine brain CT scans, MRI images have a vast array of tissue contrasts [7–9]. Due to wide spectrum of brain tumors and also different radiologic manifestations of each entity, manual process of segmentation and analyzing of brain tumors structures in various MRI images is so difficult and time consuming that usually professional neuroradiologists, or skilled neurosurgeons can perform accurately. Therefore, establishment of a reliable automatic brain tumor segmentation technique could play a

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significant role in better diagnosis and treatment of these patients. These intelligent techniques could assist radiologists and neurosurgeons in better evaluation of tumor's volume, shape and better determination of tumor's border with adjacent normal tissues. During last decade, several models have been proposed for automatic or semi-automatic determination of brain tumors in MR images, however many of them only provide brain tumor type classification or MR image segmentation, separately. Therefore, a multi-task structure method which combine these two entities at the same time would gain more desirable results in real-world scenarios. We think multi-task learning methods could have better performance when there are shared principles or common knowledge between tasks. So, in this study, we use a new deep multi-task learning method for simultaneous brain tumor segmentation and classification.

2. Methods

2.1. Data selection and study design

The brain tumor T1-weighted contrast enhanced MRI dataset was obtained from two referral hospitals (Nanfang Hospital, Guangzhou and General Hospital, Tianjing Medical University) in China which had online free accessibility and all of them were taken between 2005 and 2010 from 233 patients. This dataset consists of 3064 slices [10] and each image has a resolution of 512 × 512 with pixel size of 0.49 × 0.49 mm^2, slice thickness of 6 mm and slide gap of 1 mm. 708 meningiomas, 1426 gliomas, and 930 pituitary tumor slices (Fig. 1) were included and the tumor border with normal tissue were manually demonstrated in each slide by three experienced neuroradiologists.

A deep architecture algorithm is designed for tumor segmentation and classification (Fig. 2). This algorithm receives a two-dimensional MR image, tumor mask and image label as inputs. Both tasks (segmentation and classification) share a common encoder while an extra decoder process is applied for segmentation achievement and identification of tumoral areas. Furthermore, a fully connected neural network is utilized for images classification as three categories of brain tumors mentioned previously (see Fig. 3).

In the evaluation of the classification model, we considered radiological tumor diagnosis, and notably, no histological examination results were used, although radiologic characteristics of these three types of tumors are usually diagnostic. additionally, the segmentation model, developed from contrast-enhanced T1-weighted MRI images, underwent meticulous validation against the corresponding radiological tumor volume. Importantly, no data from post-surgery tumor volumes were incorporated into the model testing process. This meticulous approach ensures that our segmentation model is tested against the radiological characteristics and volumes of tumors before any surgical interventions.

2.2. Image preprocessing

All images are normalized, and resized to 224 × 224 pixels. Reducing the dimension of images decreases the computational complexity. Different image reduction sizes are examined and best results are given by 224 × 224 pixels. Since deep learning neural networks extract the valuable features for classification and segmentation tasks automatically, we do not apply any additional filters in this stage.

2.3. Multi-task deep learning architecture

The deep multi-task learning approach is usually used for multi-task learning, including segmentation and classification of MR images. Multi-task learning can optimize performance when tasks share fundamental principles or common knowledge [11,12]. In this research, a multi-task deep learning architecture is employed to analyze different MR images by classifying them into meningiomas, gliomas, and pituitary adenomas groups, as well as segmenting them to identify the tumor mass border.

The encoder-decoder architecture is based on U-NET, which uses convolution blocks followed by skip connections to preserve spatial information without pooling. The convolutional layer applies filters to detect spatial patterns such as edges, lines, and bends in an image based on changes in intensity values. The multi-task deep architecture uses 3x3 convolution windows and increases the number of filters from 64 to 1024 in the encoder to move from low-level to high-level feature maps. At the end of the encoder process, a set of features can determine the tumor type and its location in an MR image.

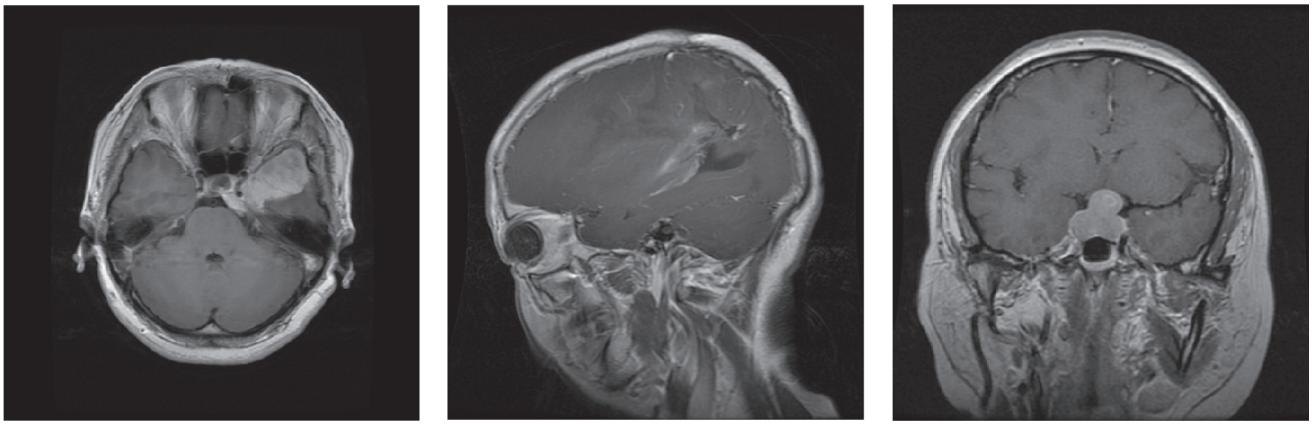
To begin decoding, a sampling layer is employed, followed by a convolution layer that reduces the number of features by a factor of 2. The up-sampled features are then merged with the corresponding level of features from the encoder section via concatenation to produce the final image output. In the segmentation phase, the output from the decoder is treated as the segmented output, and a loss function is used to evaluate its performance as follow

$$Dice = \frac{2^*|x \bigcap y| + \varepsilon}{|x| + |y| + \varepsilon} \tag{1}$$

$$Loss_{seg} = -Dice \tag{2}$$

where x and y show desired and segmented images respectively. ε is the smoothing factor and used to avoid a division by zero.

To perform the classification task, the encoder's output is combined with a convolution layer and a max-pooling layer. The result is then flattened for processing. The fully connected neural network contains



(a) (b) (c)

Fig. 1. (a) Meningiomas, (b) gliomas, (c) pituitary tumors.

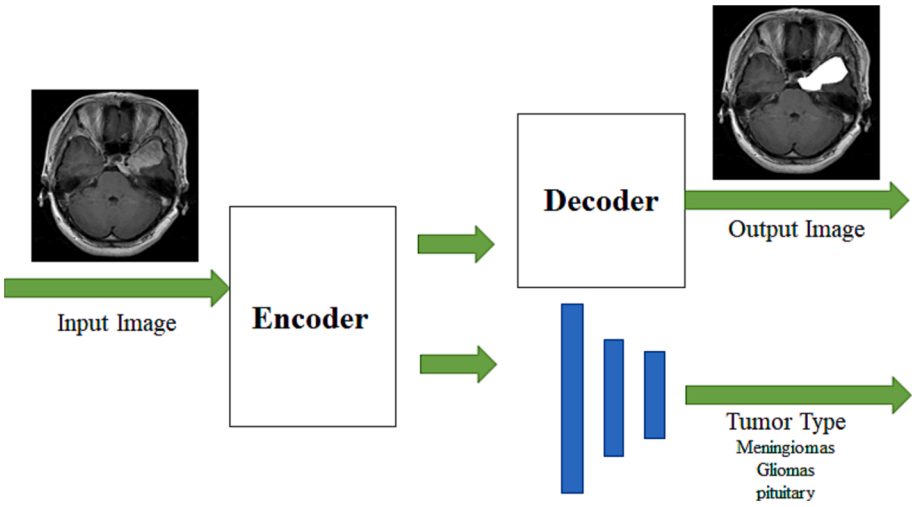


Fig. 2. The general model of the deep architecture.

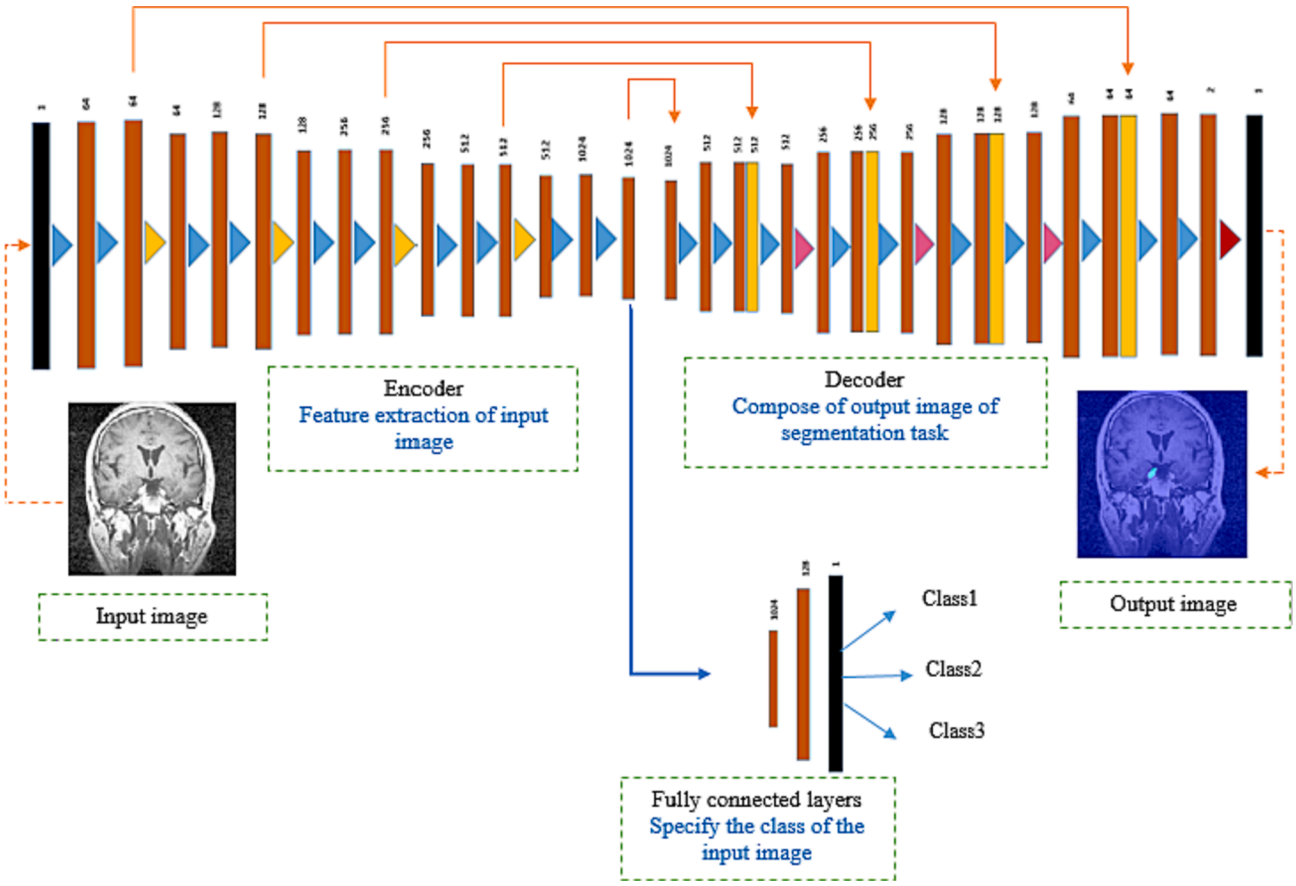


Fig. 3. The detailed architecture of encoder-decoder.

two dense layers, each with 128 and 64 neurons, respectively. The ReLU activation function is applied, and a dropout rate of 0.5 is utilized. The last layer is a dense layer with three neurons, which utilizes a sigmoid activation function to classify the images. The loss function used in this process is defined as

$$\mathrm{Loss}_{\mathrm{class}} = -\frac{1}{n}\sum_{i=1}^{n}[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)] \tag{3}$$

where n shows the number of training images, y and $\hat{y}$ indicates the actual and predicted class labels.

2.4. Performance evaluation

The model effectiveness was evaluated by calculating its performance using several metrics including dice, accuracy, sensitivity, and specificity, which were defined for the segmentation task as follow

$$Accurcy = \frac{TP + TN}{TP + FP + TN + FN} \tag{4}$$

$$sensitivity = \frac{TP}{TP + FN} \tag{5}$$

$$Specificity = \frac{TN}{TN + FP} \tag{6}$$

The true positive (TP), true negative (TN), false positive (FP), and false negative (FN) values were used to calculate the performance metrics for the segmentation task. Moreover, to measure the efficiency of the classification task, several metrics were used, including accuracy, sensitivity, specificity, and area under the receiver operating characteristic (ROC) curve (AUC).

3. Results

The proposed model was trained, validated, and tested on magnetic resonance (MR) images. The dataset consisted of 3064 images, 70% (2143 images) used for training and 30% (921 images) used for testing. During the 50-epoch training phase, the model was trained with a mini-batch size of 9 and a learning rate of 5e-4 using the Adam optimizer to minimize the loss function. Evaluation metrics, including accuracy, sensitivity, specificity, area under the curve (AUC), and dice, were recorded and presented in Tables 1 & 2. To assess the impact of the multi-task structure on the model's performance, the second row of each table shows the results of each task using only the single-task architecture. For the classification task, the model achieved an accuracy of 0.97, sensitivity of 0.95, specificity of 0.97, and AUC of 0.96 with the multi-task architecture, compared to an accuracy of 0.96, sensitivity of 0.94, specificity of 0.96, and AUC of 0.94 with the single-task architecture. Similarly, for the segmentation task, the model achieved an accuracy of 0.97, sensitivity of 0.97, specificity of 0.91, and dice of 0.92 with the multi-task architecture, compared to an accuracy of 0.91, sensitivity of 0.95, specificity of 0.90, and dice of 0.90 with the single-task architecture. These results suggest that the multi-task structure improves the evaluation metrics for both tasks. We conducted Student's *t*-test for each structure in both classification and segmentation tasks (Tables 3 & 4). The t-value for classification was 34.47 with a very small p-value (0.000), indicating significant difference in mean accuracy between multi-task and single-task structures. For segmentation, the t-value was 45.44 with a p-value of 0.001, showing that the multi-task structure outperformed the single-task structure significantly. Therefore, we could conclude the multi-task structure is superior for both classification and segmentation tasks.

Table 2 presents the results of the segmentation task for two different methods: Multi-Task U-Net and Single-Task U-Net. The Dice scores are reported as mean dice ± standard deviation. These Dice scores indicate the level of agreement between the predicted and ground truth segmentation masks. A higher Dice score suggests better segmentation performance. The segmentation task involves distinguishing between two classes: normal tissue and tumor. The reported Dice scores are the averages across these classes. For Multi-Task U-Net, the average Dice score is 0.92 with a very small standard deviation of 0.001, indicating high consistency in segmentation performance. Single-Task U-Net, while still performing well with a mean Dice score of 0.90, shows a slightly higher variability, as reflected in the larger standard deviation of 0.012. In comparison, the results suggest that the Multi-Task U-Net outperforms the Single-Task U-Net in the segmentation task. The higher mean Dice score of 0.92 for Multi-Task U-Net, as opposed to 0.90 for Single-Task U-Net, indicates a superior ability to accurately segment both normal tissue and tumor regions. Additionally, the smaller standard deviation in the Multi-Task U-Net's Dice score (0.001) implies more consistent and robust segmentation results across different instances. Therefore, these findings suggest that the Multi-Task approach enhances the segmentation performance compared to the Single-Task approach.

To provide further comparison, we utilized two popular deep learning architectures, AlexNet and GoogleNet, in our classification task (Table 5). Our results demonstrate that the proposed architecture achieved superior performance with an accuracy of 0.97, sensitivity of 0.95, and specificity of 0.97 compared to AlexNet (with accuracy of 0.94, sensitivity of 0.94, and specificity of 0.95) and GoogleNet (with accuracy of 0.92, sensitivity of 0.96, and specificity of 0.96). Notably, our findings suggest that the proposed architecture achieved better results with a lower level of complexity. We conducted Student's *t*-test analysis to compare the classification accuracy of these models (Table 6). The t-value for AlexNet and GoogleNet comparison was 31.33 with a p-value of 0.001, indicating a significant difference in performance between these two networks. The t-value for GoogleNet and Proposed model comparison was 46.93 with a p-value of 0.003, also indicating a significant difference in performance. The t-value for AlexNet and Proposed model comparison was 54.68 with a p-value of 0.008, again indicating a significant difference in performance. Overall, the Proposed

Table 1
Result of classification task.

Method	Class	Accuracy	Sensitivity	Specificity	AUC
Multi-Task U-Net	Meningiomas	0.97 ± 0.001	0.95 ± 0.004	0.97 ± 0.020	0.95 ± 0.001
	Gliomas	0.97 ± 0.003	0.95 ± 0.001	0.97 ± 0.001	0.96 ± 0.001
	Pituitary	0.97 ± 0.010	0.97 ± 0.002	0.98 ± 0.010	0.96 ± 0.002
	Average	0.97 ± 0.004	0.95 ± 0.002	0.97 ± 0.010	0.96 ± 0.001
Single-Task U-Net	Average	0.96 ± 0.002	0.94 ± 0.003	0.96 ± 0.014	0.94 ± 0.02

Table 2
Result of segmentation task.

Method	Accuracy	Sensitivity	Specificity	Dice
Multi-Task U-Net	0.97 ± 0.001	0.97 ± 0.002	0.91 ± 0.001	0.92 ± 0.001
Single-Task U-Net	0.91 ± 0.001	0.95 ± 0.003	0.90 ± 0.002	0.90 ± 0.012

Table 3
Statistical evaluation of classification task.

	Multi-task Structure	Single-task Structure
Mean of Accuracy	0.97	0.96
Std of Accuracy	0.004	0.002
t-value	34.47	
p-value	0.000	

Table 4
Statistical evaluation of segmentation task.

	Multi-task Structure	Single-task Structure
Mean of Accuracy	0.97	0.91
Std of Accuracy	0.001	0.001
t-value	45.44	
p-value	0.001	

Table 5
Comparison of proposed Multi-Task structure with pretrained networks in classification task.

Network	Accuracy	Sensitivity	Specificity
AlexNet	0.94	0.94	0.95
GoogleNet	0.92	0.96	0.96
Proposed	0.97	0.95	0.97

Table 6
Statistical test of proposed Multi-Task structure with pretrained networks in classification task.

Network vs. Network	t-value	p-value
AlexNet - GoogleNet	31.33	0.001
GoogleNet - Proposed	46.93	0.003
AlexNet - Proposed	54.68	0.008

network outperformed the other two networks in terms of accuracy, and the *t*-test results suggest that these differences in performance were statistically significant.

In Fig. 4, the confusion matrix for the multi-task deep architecture is presented. Part (a) of the figure shows that out of 307 meningiomas, 294 were accurately classified (resulting in an accuracy of 95.76%) and 13 were misclassified. Similarly, 300 of 307 gliomas were correctly classified (resulting in an accuracy of 97.72%), while 7 cases were misclassified. Finally, for pituitary tumors, 292 of 307 were classified correctly (resulting in an accuracy of 95.11%), while 15 were misclassified. Part (b) of the confusion matrix demonstrates that the proposed architecture can accurately segment MR images, with a correct classification rate of 93% for normal parts and 4.5% for tumor parts. The model also showed a low misclassification rate of only 2% of normal parts as tumor and 0.5% of tumor parts as normal, based on the test

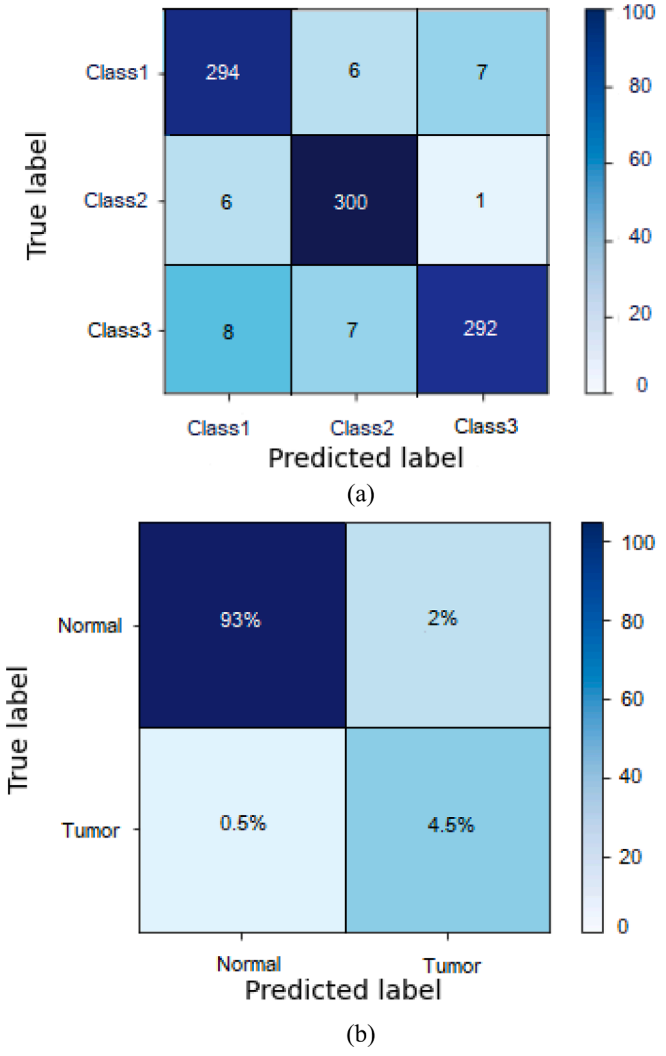


Fig. 4. Confusion matrix of deep multi-task architecture (a) confusion matrix of classification task where class 1, class 2 and class 3 shows meningiomas, gliomas, pituitary tumors respectively, (b) confusion matrix of segmentation task.

dataset.

Fig. 5 depicts the ROC curve and AUC value for each of the three classes in the classification task: meningiomas, gliomas, and pituitary tumors, achieving AUC values of 0.959, 0.964, and 0.967, respectively. These results indicate that the model is valid for the classification and segmentation of these brain tumors in MR images and performs adequately on new MR images and the proposed model can accurately differentiate tumor borders in MR images (Fig. 6).

4. Discussion

The wide range of differences between the tumor characteristics and its normal surrounding tissue such as size, location, shape, tumor heterogenicity and tumor border irregularity could complicate the segmentation process in medical images processing. Several methods have been proposed for accurate and efficient identification of brain tumors in MR images. These algorithms could be classified as three distinct categories:

(1) Machine learning approaches: A hand-crafted or predefined features are usually used in these methods. A feature extraction algorithm extracts the key information of the image and then a discriminative model (like SVM, random Forest and fuzzy clustering) is trained to demonstrate tumor in normal surrounding tissue [13]. The process of extraction of features, edge-related details and other essential information is so time-consuming. In the other hands, these methods have poor performance in segmentation of tumors with infiltrative nature or irregular borders.
(2) Multi-atlas registration algorithms: These methods registrate and label fuse so many normal brain atlases to a new image modality which require high speed hardware for the need of large number of atlases [14].
(3) Deep learning methods: These methods extract crucial features for segmentation or classification of brain tumors which have established remarkable improvements in this field [15–17].

A brain tumor segmentation network by exploring both task-modeling structure and task-task structure had acceptable results [18,19]. The first proposed network explored task structure and uses a novel approach to improve tumor detection accuracy from multi-modality MRI images, However, the model showed failure in cases where there was an identical contrast between the tumor area and the background. While the other one utilized cross-modality deep feature learning to improve accuracy by extracting and combining features from two different types of imaging modalities. The potential limitation of this approach was that the network architectures of the modal generator and the segmentation predictor needed to be similar, which could limit its flexibility in accommodating different modalities or tasks. Also, both networks had limitations of small size of data for training, and high computational complexity. It is important to consider these limitations when applying these methods in practice.

A one-pass multi-task network (OM-Net) used shared and task specific parameters to learn discriminative and joint features [20], both learning-based training and online training data transfer approaches were applied for tumor segmentation. Use of shared and task-specific parameters resulted in a significant improvement in brain tumor segmentation compared to other models. However, it was not sufficient to evaluate the performance of a model based on only one dataset which could be challenge full to obtain due to limited availability of diverse brain imaging data.

The parliamentary decision-making concept was used in an assembly Net model in order to solve the problem of limited training data in brain MRI [21]. A majority voting among neighboring U-Nets was applied to overcome the imbalanced data problem caused by small size of tumors volume compared to the rest of brain. Moreover, use of a large ensemble of models reduced overfitting and increased the generalization ability of

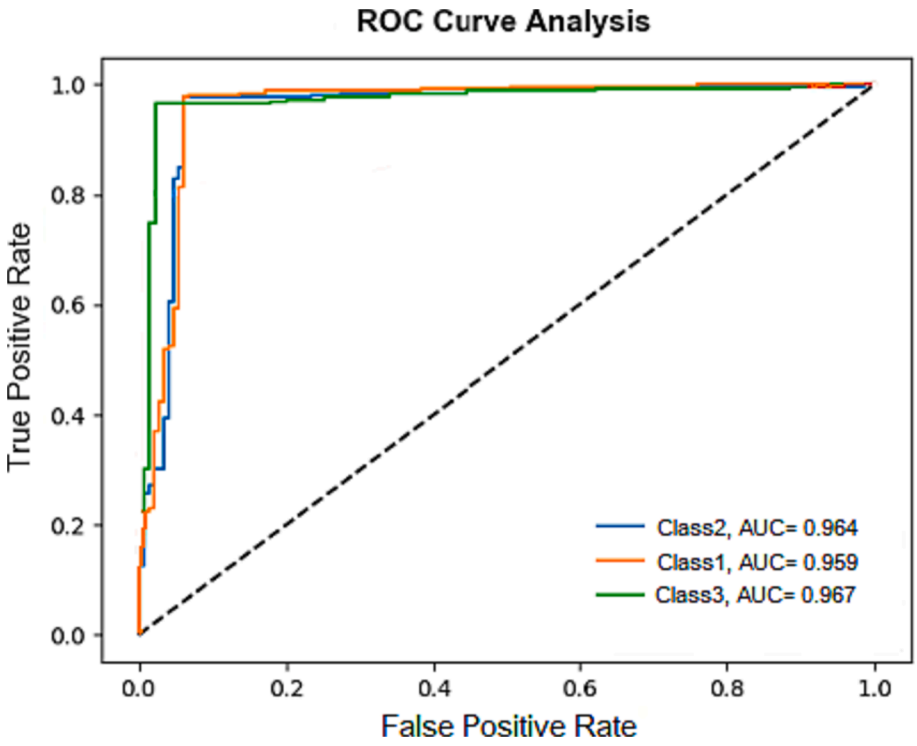


Fig. 5. The ROC curve of deep multi-task architecture.

the network. AssemblyNet also employed a self-adaptive strategy that adjusted the weighting of each model in the ensemble during training, which further enhanced its performance. However, this model had some limitations and challenges that should be taken into account. One of the main limitations was the high computational cost of training and inference due to the large ensemble of CNNs. This could make the method less feasible for some low-resource settings or real-time applications. Another limitation was the lack of interpretability of the model, which could hinder the understanding of the underlying mechanisms of segmentation process. Finally, although AssemblyNet has shown promising results on various datasets, it may not generalize well to new unseen data or different imaging modalities.

In another study, a fully automatic brain tumor segmentation based on convolutional neural network (CNN) revealed much better results in compare with previous state-of-the-art techniques [22]. Although the model could avoid a lot of repeated calculations, but the use of GAN models for post-processing the outputs of a segmentation model could have some drawbacks. One potential issue is that the post-processing may introduce artifacts or errors into the final output. Additionally, the use of GANs can be computationally expensive and time-consuming and the GAN model may require a large amount of training data to produce accurate results, which may not always be available. Finally, the performance of the GAN model might be highly dependent on the specific chosen architecture and hyperparameters, which could make it difficult to generalize to other datasets or applications.

Another innovative study [23] was conducted by proposing a pre-processing approach which focuses on a small part of image to decrease computing time and prevent overfitting on in a cascade deep learning model had optimal results in image segmentation. Their model achieved competitive results on the BRATS 2018 dataset, with mean dice scores of 0.9203 for the whole tumor, 0.9113 for the enhancing tumor, and 0.8726 for the tumor core. The model reduced computational time and enables fast prediction of clinical images by removing insignificant pixels. However, the model had limitations when dealing with tumor volumes larger than one-third of the whole brain. This was due to the fact that an increase in the size of the expected tumor area can result in a decrease in the performance of feature extraction.

An augmented tumor region obtained by dilation as the region of interest instead of the original tumor region was used to improve brain tumor classification performance [24]. This augmented region included surrounding tissues that could provide important information for tumor types. Additionally, the augmented region was divided into fine ring-form subregions to increase classification accuracy. This approach had several advantages. First, using the augmented tumor region as the ROI instead of the original tumor region improved the performance significantly because tumor-surrounding tissues characteristic could discriminative between different tumor categories in many cases. Second, the ring-form partition method compensated for the loss of spatial information and made the final concatenated feature representation more discriminative. However, this approach also had some limitations. For example, it required a pre-segmented tumor region for training and testing, which might limit its applicability in some scenarios. Additionally, the ring-form partition method might not be effective for tumors with irregular shapes or for tumors located in the periphery of the brain.

In another attempt, convolutional neural networks (CNNs) was used to segment brain tumors and analyze quantitative features such as shape, texture, and signal intensity for predicting clinical outcomes like survival and response to therapy [25]. The benefits of this method include reduction in need for manual interpretation of medical images, higher rate of speed and accuracy of disease detection and identification in medical images, increasing the accuracy of microscopic image detection and interpretation, and the ability to predict and anticipate diseases based on quantitative image features. However, this method also has some limitations, such as the need for large and diverse data for training, difficulties in using machine learning algorithms in analyzing medical images with heterogeneous features, and problems such as overfitting and generality in predicting clinical diseases.

And finally, a two-part segmentation network called DeepSeg, consisting of a CNN-based encoder for spatial information extraction and a decoder for full-resolution probability mapping was developed [26] and evaluated successfully on MRI datasets of brain tumor segmentation

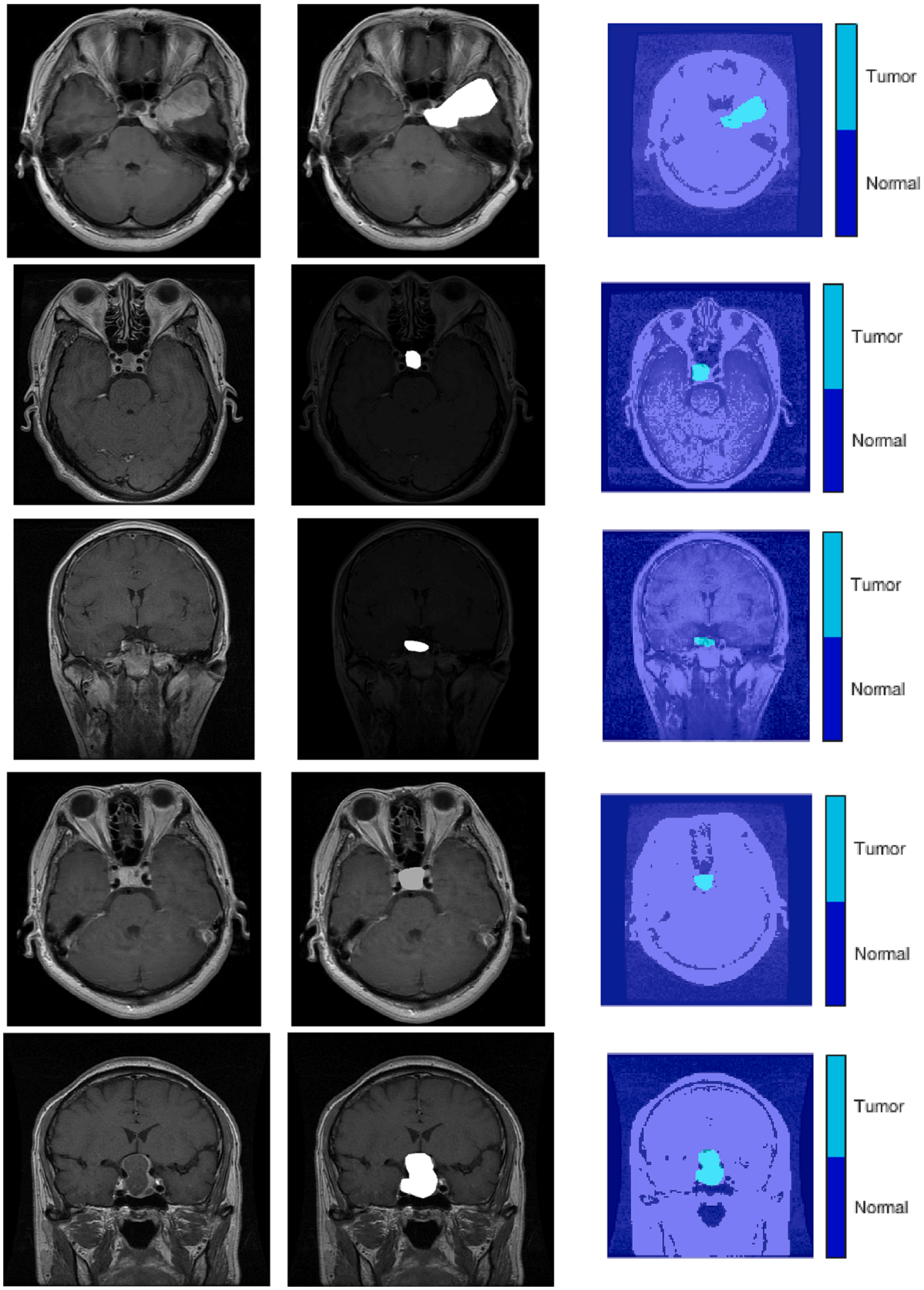


Fig. 6. Brain tumor segmentation results, images in (a) shows the original MR images, (b) hand-segmented tumor images, and (c) the results of proposed architecture.

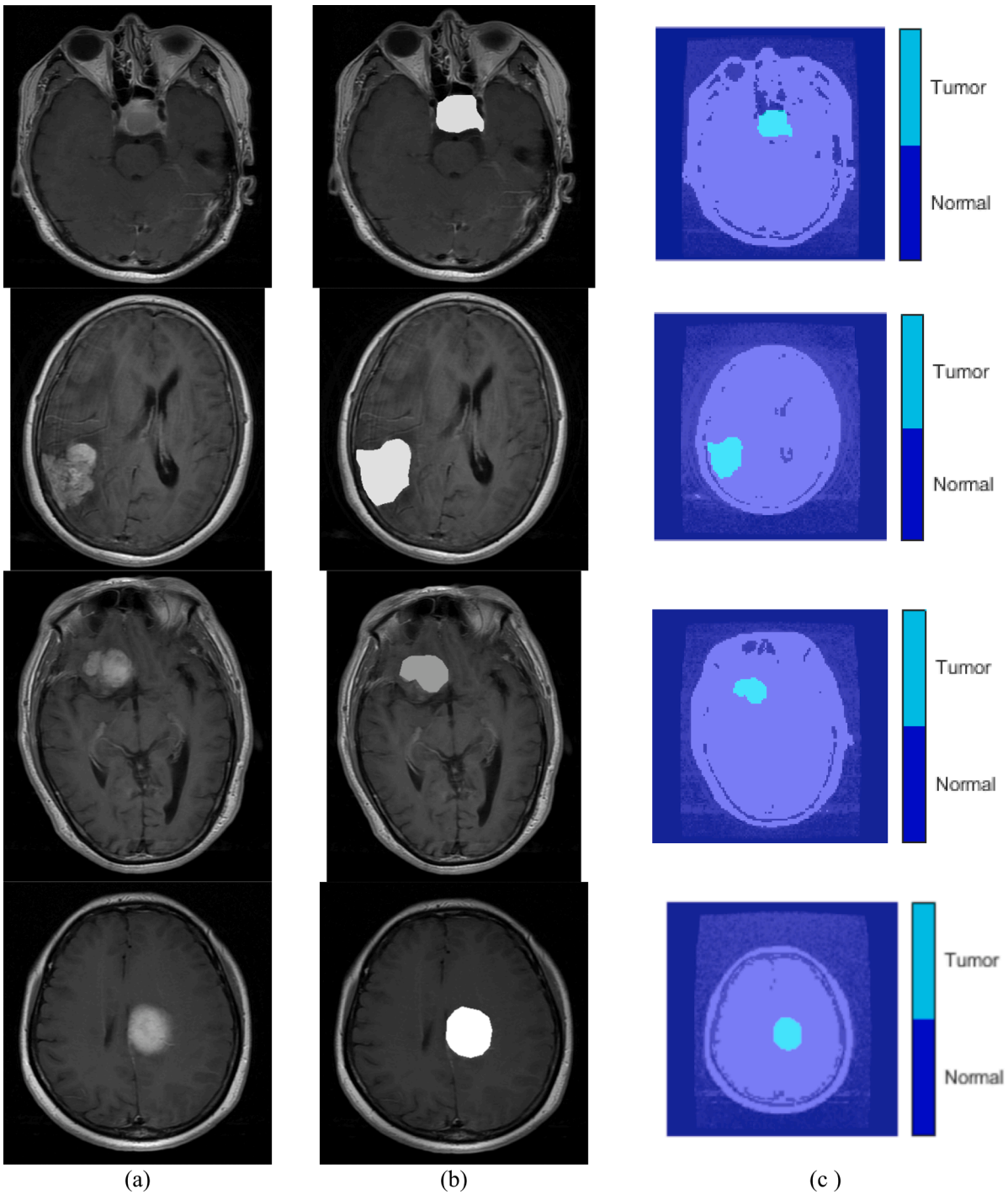


Fig. 6. (*continued*).

(BRATS 2019) challenge. Interestingly, this method investigates the effect of different deep learning models in feature extraction. However, some of these models may have a large number of layers, which can lead to overfitting and require high memory systems for data processing. This may limit the practical use of this model in clinics which have minimal access to such systems.

Here, we introduce a deep multi-task learning method for simultaneous segmentation and classification of 3 major subtypes of supratentorial brain tumors in MR images. As there are shared principles or common knowledge between proposed tasks, we think multi-task learning could gain higher levels of performance. Our method could gain acceptable performance in segmentation of brain tumors which significantly has better results than single-task architecture. This superiority is also existed in classification of tumors which suggest that the multi-task structure improves the evaluation metrics of both classification and segmentation tasks. In the proposed multi-task structure, a common model is used to learn two tasks simultaneously, which offers advantages such as improved data performance, reduced overfitting through shared representations, and faster learning. The model outperforms state-of-the-art methods in terms of evaluation metrics and has the potential to be used as a primary screening tool to assist primary service staff in referring suspected patients to specialists. However, the primary constraint of this study is the requirement for the input dataset to be used for both segmentation and classification tasks. It is not

feasible to use images solely for one task or the other, as the model has a multi-task architecture with a shared encoder. Therefore, each image must have both mask and label information available simultaneously. Nevertheless, there are limited studies in this area due to structural weaknesses and a lack of testing on comprehensive datasets, which include the absence of appropriate encoders and decoders, as well as suitable classification methods.

5. Conclusion

We propose a deep multi-task learning approach for automatic diagnostic diagnosis of 3 common supratentorial primary brain tumors (meningiomas, gliomas and pituitary adenomas) in contrast enhanced T1 weighted MR images. As a novel method, simultaneous segmentation (automatic identification of tumoral border) and classification (automatic prediction of tumor type) was proposed and interestingly reliable and promising results were achieved in both segmentation and classification tasks.

CRediT authorship contribution statement

Shirin Kordnoori: Methodology, Software, Writing – original draft. **Maliheh Sabeti:** Conceptualization, Formal analysis, Methodology, Software, Supervision, Validation, Writing – original draft, Writing – review & editing. **M.H. Shakoor:** . **Ehsan Moradi:** Conceptualization, Formal analysis, Investigation, Validation, Visualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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